Sheet 2

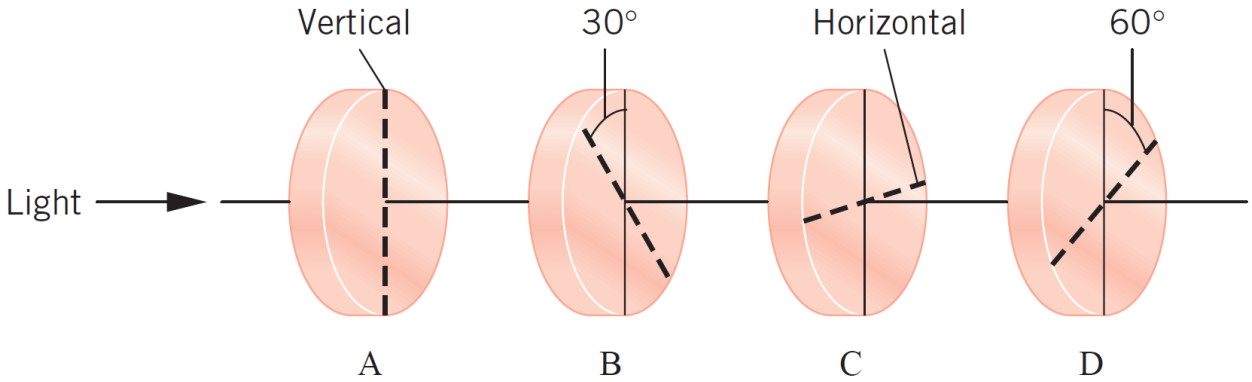
1) Unpolarized light with intensity I_o is incident on two polarizing filters. The axis of the first filter makes an angle of $60°$ with the vertical, and the axis of the second filter is horizontal. What is the intensity of the light after it has passed through the second filter?

2) Light of intensity I_o and polarized parallel to the transmission axis of a polarizer is incident on an analyzer.

   a) If the transmission axis of the analyzer makes an angle of $45°$ with the axis of the polarizer, what is the intensity of the transmitted light?

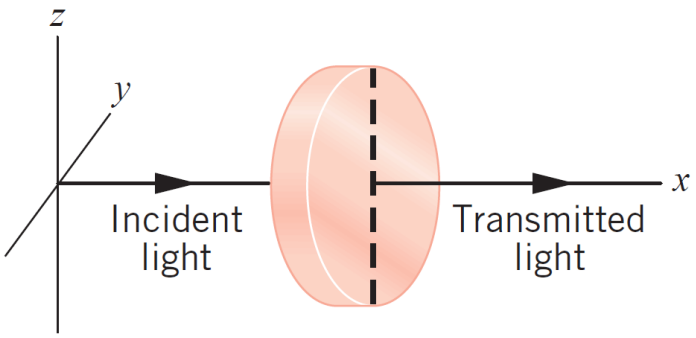
   b) What should the angle between the transmission axes be to make $I/I_o = 1/3$?

3) The drawing shows four sheets of polarizing material, each with its transmission axis oriented differently. Light that is polarized in the vertical direction is incident from the left and has an average intensity of $27\ \mathrm{W/m^2}$. Determine the average intensity of the light that emerges on the right in the drawing when sheet A alone is removed, when sheet B alone is removed, when sheet C alone is removed, and when sheet D alone is removed.



4) The drawing shows light incident on a polarizer whose transmission axis is parallel to the z axis. The polarizer is rotated clockwise through an angle α. The average intensity of the incident light is $7.0\ \mathrm{W/m^2}$. Determine the average intensity of the transmitted light for each of the six cases shown in the table.

	Intensity of Transmitted Light	
Incident Light	$\alpha = 0°$	$\alpha = 35°$
(a) Unpolarized		
(b) Polarized parallel to z axis		
(c) Polarized parallel to y axis		



5) A parallel beam of unpolarized light in air is incident at an angle of $54.5°$ (with respect to the normal) on a plane glass surface. The reflected beam is completely linearly polarized.

a) What is the refractive index of the glass?

b) What is the angle of refraction of the transmitted beam?